

National Environmental Data Management Plan

SECTION THREE: Global, Regional and National Frameworks

Benchmarking Best Practices in Environmental Data Governance

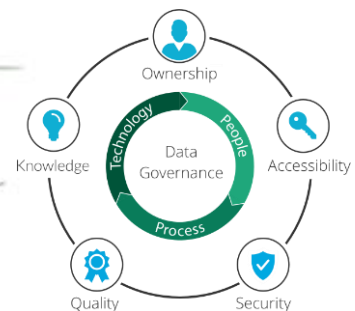


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List of Acronyms

Acronym	Meaning
ACSAD	Arab Center for the Studies of Arid Zones and Dry Lands
ALECSO	Arab League Educational, Cultural and Scientific Organization
CEDARE	Centre for Environment and Development for the Arab Region and Europe
DBMS	Database Management System
DSD	Data Structure Definitions
EDS	Environmental Data Service (UK)
EEA	European Environment Agency
EIS	Environmental Information System
ENPI	European Neighbourhood Partnership Instrument
ENPI-SEIS	European Neighbourhood Partnership Instrument – Shared Environmental Information System
ESA	European Space Agency
ESCWA	Economic and Social Commission for Western Asia
EU	European Union
FAIR	Findable, Accessible, Interoperable, Reusable
FDES	Framework for the Development of Environment Statistics
GEDS	Global Environmental Data Strategy
GIS	Geographic Information System
GRID	Global Resource Information Database
IBM	International Business Machines Corporation
INSPIRE	Infrastructure for Spatial Information in the European Community
ISO	International Organization for Standardization
IT	Information Technology
IV&V	Independent Verification and Validation
LAS	League of Arab States
MEA	Multilateral Environmental Agreement
MEIC	Ministry of Environment Information Centre (Macedonia)
MEPP	Ministry of Environment and Physical Planning (Macedonia)
MIT	Council for Management and Development of Business Solutions (Sweden)

NASA	National Aeronautics and Space Administration
NEDMP	National Environmental Data Management Plan (Kuwait)
NERC	Natural Environment Research Council (UK)
PKI	Public Key Infrastructure
SDG	Sustainable Development Goal
SDI	Sustainable Development Indicator
SDMX	Statistical Data and Metadata Exchange
SEEA	System of Environmental-Economic Accounting
SEIS	Shared Environmental Information System
SNA	System of National Accounts
UK	United Kingdom
UKRI	UK Research and Innovation
UN	United Nations
UNEA	United Nations Environment Assembly
UNEP	United Nations Environment Programme
UNGA	United Nations General Assembly
UNSD	United Nations Statistics Division
WCMC	World Conservation Monitoring Centre
WESR	World Environment Situation Room
XML	Extensible Markup Language

1 Introduction

The development of a sound National Environmental Data Management Plan (NEDMP) for Kuwait cannot proceed in isolation of the global practices. Overcoming decades of efforts made by various Governments, United Nations Organizations and Regional Organizations worldwide to establish environmental data governance, management systems, frameworks and structures at the global, regional and national level demands assessment of global and regional initiatives as well as of national experiences in other countries. The United Nations Environment Programme (UNEP) Global Environmental Data Strategy (GEDS) and statistical data frameworks, the regional example of the European Environment Agency (EEA) shared Environment Information System (SEIS) in Mediterranean region and Southeast Europe, and national activities carried out recently in the frameworks of sectors strategies in North Macedonia, as well as national practices in Sweden and in the United Kingdom can provide numerous cases of implemented practices, standards, technologies, methods and systems that could serve as examples or benchmarks when developing a National Environmental Data Management Plan in Kuwait.

The contextualisation of the NEDMP has three main purposes. First, it aims to establish a strong foundation to the Plan based on universally adopted principles, including the UN Global Environmental Data Strategy (GEDS) and the Fundamental Principles of Official Statistics. Second, it aims at aligning the NEDMP with the global and regional data systems and meeting the international obligations resulting from the accession to multilateral environmental agreements and the achievement of the Sustainable Development Goals. Third, it aims at integrating Kuwait into the international system of environmental data, and thereby enhancing the interoperability and coordination with UN organizations and other countries in the region and abroad.

This chapter offers a review of some international, regional and national practices for environmental data management. It focuses on the main objectives, characteristics of the governance systems and activities that were implemented, while drawing the links with the five strategic objectives of the NEDMP. A comparison between the best international practices and national practices with the NEDMP's strategic objectives reveals that in terms of the global transition to data being trusted, shared, comprehensive, shared, integrated and accessible; Kuwait NEDMP is fully aligned and thus working in line with the global environmental data management principles for sustainable development.

2 Conceptual Clarification of Environmental Data Governance Framework Components and Terminologies

The building blocks interrelationship of an environmental data governance framework should be clearly defined, while exploring the unique roles that each component plays in a holistic data governance framework.

Environmental Data Strategy at the highest level defines the long-term vision and strategic priorities of how environmental data will be used to be a valuable resource for both policy and operational decision making and overall institutional performance.

Environmental Data Governance, establishes the rules, policies, roles and accountabilities to effectively manage environmental data and ensure that it is accurate, complete, consistent and secure.

Environmental Data Management Plan is an instance of an implementation of the strategic and governance elements previously identified. It provides a description of the activities required to manage the data for the duration of the plan or observational activities including data capture, storage, verification and sharing.

Data Implementation Plan is a detailed plan for activities required to implement activities as detailed in the Data Management Plan. It includes such information as specific systems and infrastructure required, in addition to details of associated activities.

Data Action Plan lists specific activities that need to be carried out within a given timeframe and assigns roles to individuals to ensure the implementation of the plan is tangible, measurable and result oriented.

When those terms joined together, they form a logical and consistent framework from intent to action and with appropriate definitions of their respective responsibilities.

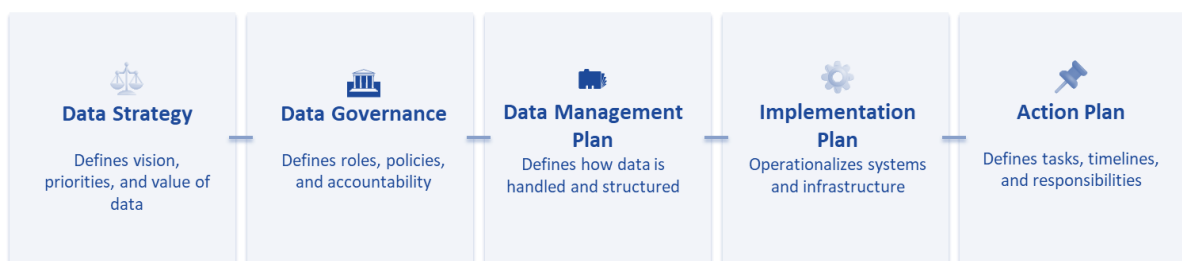


Figure 1 Data Governance Framework Conceptual Flow

Table 1 Core Focus of Data Governance Framework Components

Term	Core Focus
Data Strategy	"Why": High-level choices to achieve business goals from data
Data Governance	"Who, What, When, Where": Roles, policies, and authority over data
Management Plan	Project lifecycle: How data is handled from creation to preservation
Implementation Plan	Deployment: How to transition from design to operational system
Data Action Plan	Execution: Concrete steps with responsibilities and timelines

3 International, Regional and National Best Practices in Environmental Data Management

Effective management of the environment requires strong data governance to support the evidence-based decision-making required for environmental management, climate change policy and sustainable development planning. A growing number of governments, international and regional organizations at the global, regional and national levels are implementing environmental data management plans and digital environmental information systems to collate, process and disseminate large quantities of environmental data from existing national observation networks, satellite images and national statistical sources.

Benchmarking international, regional and national environmental data strategies and plans is a key milestone in preparation of work on the development of the Kuwait National Environmental Data Management Platform (NEDMP). This process is essential to ensure that the development of the NEDMP is built on robust basis and in accordance with the highest international standards. Countries, and international and regional organizations have been working for decades to establish mechanisms and practices for collection, provision, validation and dissemination of environmental data. The UNEP Global Environmental Data Strategy, the EU INSPIRE Directive, and national environmental data strategies and initiatives in countries such as Singapore, Brazil and the United States of America, etc. are examples of such practices.

The United States Environmental Protection Agency uses management systems to make sure the information they collect about the environment is reliable. Some countries like Rwanda and Singapore have come up with their ways to deal with environmental problems that are specific to their areas. These local approaches can be used in places

which shows that they can make a big difference in efforts to protect the environment worldwide.

Even though we have made some progress managing environmental information is still a challenge. There are problems with the quality of the information getting it to work with business systems and concerns about who controls the information. We need to solve these problems to make monitoring and reporting better which is necessary for meeting sustainability goals in our own country and, around the world.

Benchmarking key environmental data strategies and plans will allow to identify, extract and transplant practices, mechanisms and tools in addition to adopting and using standards and governance practices that are internationally recognized, thereby facilitating the planning related to the NEDMP, ensuring that the NEDMP is compatible with and with the global and regional data systems. The following is not a comprehensive review of all the global environmental data management systems but rather highlighting key examples of some of the most relevant best practices and case studies. In this regard, the state of Kuwait will be able to meet its international reporting obligations, achieve regional coordination and interoperability, and exploit the latest developments in the field of information and communication technologies such as artificial intelligence. In summary, benchmarking key international, regional and national environmental data strategies and plans will provide a strategic basis for the NEDMP that enhances its reliability, effectiveness and sustainability.

Main critical characteristic of evaluating the efficiency of a practice is to measure the degree to which this intervention has responded to environmental challenges. This evaluation criteria categorises to what extent the case study has successfully achieved the objectives. The process entails analysing the interventions carried out in response to environmental issues taking into consideration their scope, effectiveness, and any unplanned consequences. Applying evidence from monitoring and evaluation process is important to compare interventions against its predefined objectives and benchmarks, thereby enabling an inclusive understanding of success aspects, including funding, governance, and community engagement.

3.1.1 The Global Environmental Data Strategy (GEDS)

The implementation approach for this strategy recognizes that UNEP plays a catalytic role as a facilitator, convenor and a provider of technical tools and services to assist partners in their implementation efforts (UNEP, 2025). It is expected that governments, multilateral environmental agreements, international scientific organizations, business, civil society and Indigenous Peoples will collaborate to achieve the strategy's vision, and its seven strategic intervention points (by 2030) (UNEP, 2025). Work under this programme of work

was expected to be implemented in a gradual, adaptive and country-oriented manner with sound bases established during the first decade of the millennium (UNEP, 2025).

Strategy Mandate: Developed in response to UNEA resolution 4/23, the Global Environmental Data Strategy (GEDS) provides a foundational platform to improve the production of environmental data, how data is managed, and used worldwide (UNEP, 2019). The main objective is to address the fragmentation, inaccessibility, and poor interoperability that currently hinder effective environmental action (UNEP, 2025).

Strategy Vision: *“A trusted, inclusive, accessible and interoperable environmental data ecosystem that enables collective action at the speed and scale needed to solve the environmental crisis and support sustainable economies and societies worldwide” (UNEP, 2025, p. 5).*

GEDS Main Principles: GEDS is a voluntary initiative and respects national data sovereignty, laws, and circumstances (UNEP, 2025). It does not create new reporting obligations or impose standards. Instead, it acts as an enabling framework to reduce fragmentation, lower costs, and maximize the value of existing data (UNEP, 2025).

Table 2 GEDS Five Foundational Pillars

Pillar	Focus
Management	Policies, institutions, and practices that ensure coordinated, ethical, and sustainable data governance
Quality & Provenance	Tools and processes to ensure data are accurate, traceable, validated, and fit for purpose
Interoperability	Alignment of data formats, semantics, and machine-to-machine interfaces to enable seamless integration
Accessibility & Affordability	Equitable, secure, and sustainable access to data, with transparent licensing and user-centered design
Capacity-Building	Developing human skills, institutional frameworks, and technical infrastructure to use data effectively

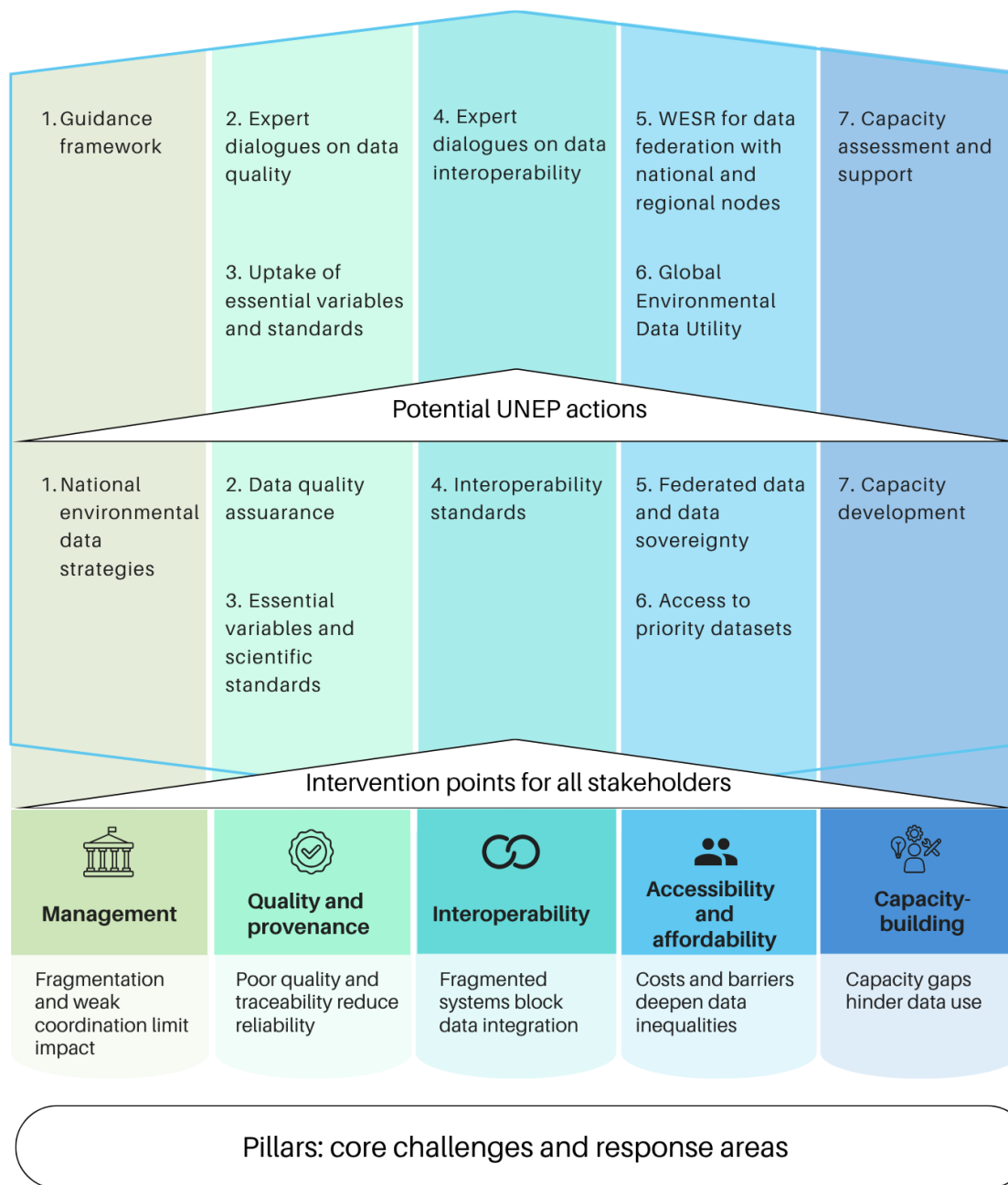


Figure 2 The Global Environmental Data Strategy (GEDS)

3.1.2 Environmental Data Management Statistical Frameworks

Environmental data management refers to the set of mechanisms, policy, standards or processes that are established to manage the acquisition, processing and dissemination of environmental data. According to the United Nations Environment Programme (UNEP), the purpose of environmental data management is to ‘ensure that high-quality, accessible environmental data is available to underpin global, regional and national responses to the

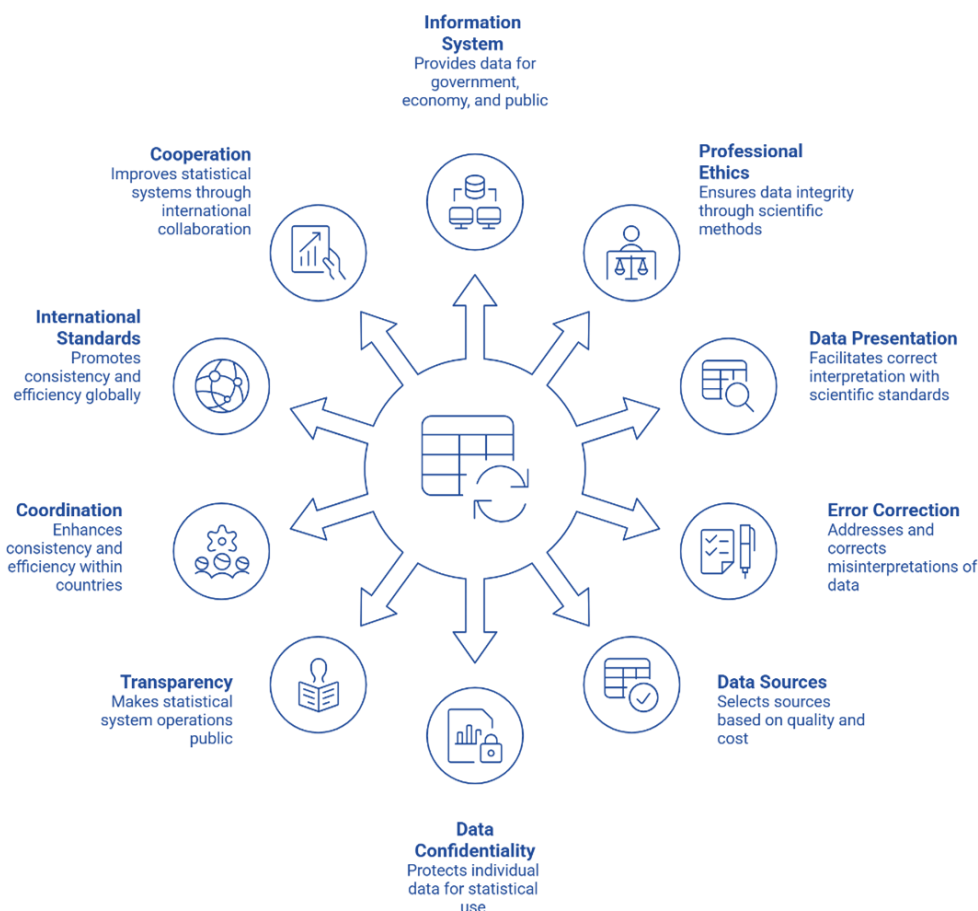
triple planetary environmental crises of climate change, pollution and biodiversity loss' (UNEP, 2023, p. 5). The United Nations Environment Programme's Global Environmental Data Strategy (GEDS) achieves this purpose, as mentioned in previous sections, through five key focus areas: i) improve data quality and sources, ii) strengthen environmental data governance, iii) promote data interoperability, iv) ensure equitable and open data access, and v) support capacity development in UN Member States (UNEP, 2023).

In order to organize global frameworks done by various actors, the United Nations Statistics Division (UNSD) published in 2013 the Framework for the Development of Environment Statistics (FDES 2013). The framework is considered the main foundation for environment statistics and socio-economic data, and an essential tool for the Environment Statistics. It is also considered an Environment Statistics Self-Assessment Tool that are designed to guide countries in assessing and enhancing their national environment programmes (United Nations, 2017).

The System of Environmental Economic Accounts (SEEA) is an internationally recognized standard for environment statistics. This system combines environmental and socio-economic data to highlight the interrelationships between the economy and the environment, using an accounting structure based on the System of National Accounts (United Nations et al., 2014). The SEEA includes the Data Structure Definitions (DSDs) for priority accounts such as air emissions, energy and water and derived indicators used for monitoring the Sustainable Development Goals (SDGs) (United Nations, 2021).

The UN Fundamental Principles of Official Statistics list 10 key principles that official statistics must be based on (UN Economic and Social Council, 2013).

Among other things, official statistics, including environmental data, shall be based on official sources and give an accurate picture of reality, compiled in a professional and objective means and that sources and methods be clearly explained so that the data can be interpreted correctly, that personal data be treated with the utmost care and not used for purposes other than statistical, and that the requirements of users shall be properly taken into account and that all users shall have an impartial and equal access to official statistics.



3.1.3 World Environment Situation Room (WESR)

The UN Environment Programme (UNEP) World Environment Situation Room (WESR) is a data, information and knowledge system functioning as a “one stop shop” for primary environmental data. UNEP's mandate is “to keep under review” the world environment situation (UNGA 239/24), and the World Environment Situation Room is one of UNEP’s key tools developed to materialize this mandate as part of the global environmental data strategy. According to the UNEP World Environment Situation Room (UNEP, 2024), the platform serves as a joined system providing access to Earth observation, thematic, and statistical environmental data to shed light on critical issues such as air pollution, biodiversity loss, climate change, pollution and water pollution among others. It aims to facilitate data-informed analysis, decision-making for UN, governments and society and to enhance environmental governance, increase transparency and take environmental progress towards the achievement of the environmental components of the Sustainable Development Goals (SDGs) and the Multilateral Environmental Agreements (MEAs).

Realizing knowledge of the environmental situation through the development of the WESR has brought UNEP to foster partnerships and network with UNEP's GRID centres, the World Conservation Monitoring Centre (WCMC), the European Space Agency (ESA), and NASA. The development of the web-based platform includes partners from the private sector, such as Google and IBM among others. In terms of functionalities, country-specific information can be found on the dynamic country platform, accessible through an online digital library of key reports for each country. Each environmental topic has a page dedicated with key statistics.

The platform converts raw data to accessible formats, including maps, charts, graphs, and tabular data, to support environmental assessments and policy development. In addition, SDG country scorecards and foresight and early warning systems are key elements that disseminate timely authoritative, primary quality-curated information ranging from policy-making at national level, to citizens' science activities.

Core Data Types available at WESR:

Table 3 Main types of data available on WESR (UNEP, 2024; UNEP, 2025)

Data Type	Description	Examples
Geospatial & Earth Observation Data	Satellite imagery, remote sensing data, and other geographically referenced information.	- Real-time global population exposure to air pollution. - Changes in glacier mass and sea-level rise. - Land cover and forest monitoring data.
Thematic Environmental Data	Data organized by key environmental topics and aligned with the Sustainable Development Goals (SDGs).	Biodiversity, Climate Change, Pollution (air, water, land, marine), Oceans, Disasters & Conflicts, Energy, and Urban environments.
Statistical & Indicator Data	Structured statistical data, including official environmental indicators.	- SDG country scorecards and environmental indicator data. - Interactive charts, tables, and country-specific dashboards.
Near Real-Time & Modelled Data	Data from sensors and analytical models for timely analysis and forecasting.	- CO₂ atmospheric concentration data. - Data for informing near real-time analysis and future predictions on environmental factors.

3.1.4 The ENPI-SEIS (European Neighbourhood Partnership Instrument (Shared Environmental Information System))

ENPI-SEIS project was part of the European Environment Agency (EEA) action programme for 2010-2014. Environmental degradation is a serious problem that threatens not only the region's sustainability but also the EU's future. ENPI-SEIS project was targeting the European Neighbourhood region, specifically Eastern Europe, the Southern Mediterranean and the Russian Federation.

Vision

ENPI-SEIS draws its basis from the broader framework of the European Union's Shared Environmental Information System (EU SEIS). SEIS is about modernizing and streamlining the collection, dissemination and use of the environmental information required for policy development and implementation. The main objective of the SEIS project is to be able to establish an environmental information system which is:

- 1- Simplified: easy to comprehend and to work in;
- 2- Efficient: gathering minimum amount of information necessary for the operations carried out in the system, the use of technology for substituting or reducing human work to a minimum, using gathered information to facilitate a more efficient decision-making and provision of services to users;
- 3- Transparent: facilitating immediate and unimpeded access to information and intelligibility of operational procedures, thereby enabling accountability and mutual monitoring;
- 4- Distributed: gathering information as close to the sources as possible and distributing information as close to the users;
- 5- Open: providing easy and simple access for all users and stakeholders and enabling informed citizen participation (European Environment Agency, 2012).



Overall Objective

To promote the protection of the environment in the countries of the ENPI area through enhanced environmental information management capacities.

Specific Objectives

- 1- Data and information flows and environmental indicators to be used for designing and reviewing environmental policies and for monitoring in order to ensure that national, regional and international obligations are fulfilled shall be determined and developed.
- 2- Strengthen the capacity of Environment Agencies and National Statistical Offices for environmental data monitoring, collection, storage, analysis and reporting.
- 3- Set up the national and regional environmental information systems in ENP countries based on the EU SEIS principles.
- 4- Track the achievements of environmental initiatives at regional level, covering activities under the Eastern Partnership, the Union for the Mediterranean, as well as Horizon 2020.

Guiding Principles (The SEIS Principles)

The initiative was built on seven core principles to guide how environmental information should be managed:

Table 4 ENPI-SEIS Guiding Principles

Principle	Description
1. Source Proximity	Information should be managed as close as possible to its source
2. Collect Once, Share Many Times	Data collected once should be shared with others for multiple purposes
3. Fulfill Reporting Obligations	Information should be readily available to easily meet reporting requirements
4. Accessibility	Information should be easily accessible to all users
5. Comparability & Participation	Data should enable comparisons at appropriate geographic scales and facilitate citizen participation
6. Public Access & Language	Information should be fully available to the public in relevant national languages
7. Open Standards	Data management should be supported through common, free, open software standards

3.1.5 The League of Arab States (LAS) Sustainable Development Indicators Initiative

The Arab League Sustainable Development Indicators Initiative represents a regional effort to collect and harmonize environmental and sustainable development indicator frameworks throughout the Arab countries.

The initiative is coordinated by the League of Arab States (LAS) in partnership with the United Nations Environment Programme (UNEP)-Office of West Asia, the Economic and Social Commission for Western Asia (ESCWA), the Arab Center for the Studies of Arid Zones and Dry Lands (ACSAD), the Center for Environment and Development for the Arab Region and Europe (CEDARE), and the Arab League Educational, Cultural and Scientific Organization (ALECSO), this initiative seeks to strengthen evidence-based environmental policy-making and reporting across the region.

Countries agreed on a proposed set of environmental indicators aligned with the Sustainable Development Goals, drawing from the initial package of 44 indicators previously approved by the Council of Arab Ministers Responsible for the Environment (League of Arab States, 2017, p. 8).

Vision

The vision of the initiative is to achieve a reliable, harmonized and sustainable framework of environmental and sustainable development indicators in the Arab region to enable countries to report effectively on achieving the Sustainable Development Goals (SDGs) of the 2030 Agenda.

Based on this aspiration, the goal is to develop an agreed set of Arab Sustainable Development Indicators (SDIs) that can be supported and implemented through achieving a clear alignment with the global SDGs while considering national and regional variability in availability and quality of statistical data in the region. The overall ambition is to enable the development of a coordinated Arab statistical system to support and facilitate the work of countries to develop an accurate evidence-based work for their strategic policies, while achieving an integrated policy framework, and enhance countries' capacities to report to international forums as requested.

Overall Objective

To promote environmental conservation and sustainable development in the Arab region through the development and harmonization of environmental and sustainable development data and indicators that support informed policy-making, monitoring, and reporting.

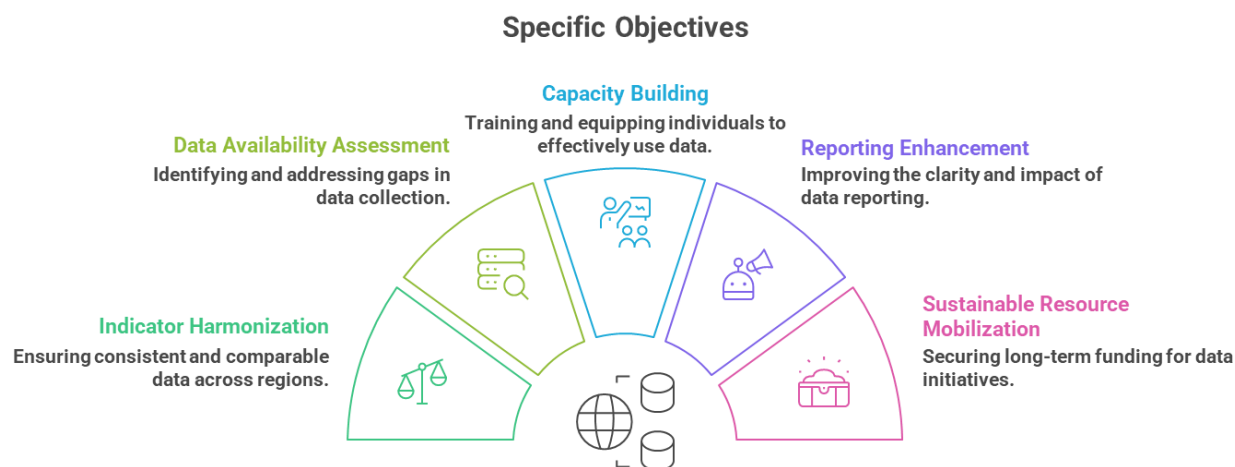


Figure 3 The Arab League Sustainable Development Indicators Specific Objectives

Guiding Principles

Table 5 Guiding Principles of Arab Initiative Derived from International Best Practices

Principle	Description
Alignment with Global Frameworks	Indicators are harmonized with the UN SDG framework and the Fundamental Principles of Official Statistics
National Ownership	Respect for national data sovereignty, legal frameworks, and country-specific circumstances
Regional Coordination	Collaboration among regional organizations (LAS, UNEP, ESCWA, CEDARE, ACSAD, ALECSO) to provide technical support.
Capacity Development	Emphasis on building sustainable human and institutional capacities rather than one-off training interventions
Transparency and Accessibility	Promotion of open access to environmental data and indicators through electronic portals and knowledge platforms

3.1.6 Republic of Macedonia: Environmental Data Management Strategy

There was an essential need to develop an Environmental Data Management Strategy for the Republic of Macedonia because the Ministry of Environment and Physical Planning (MEPP) information systems was developed through different donor projects, creating a fragmented landscape of unconnected databases. This makes environmental data in the country difficult to share, costly to assemble, and hard to verify for accuracy.

Furthermore, a "communication culture" within public administration that hinders information sharing poses a major obstacle. As noted by Jasch and Kulakov (2004, p. 23),

key success factors for the Macedonian EIS include common purpose and political support at the highest level.

Objective: This strategy, provides a framework for implementing a national environmental data management plan. Its main objective is to make consistent environmental information available to support informed rational decision-making, public access, and to be compliant with EU legislation.

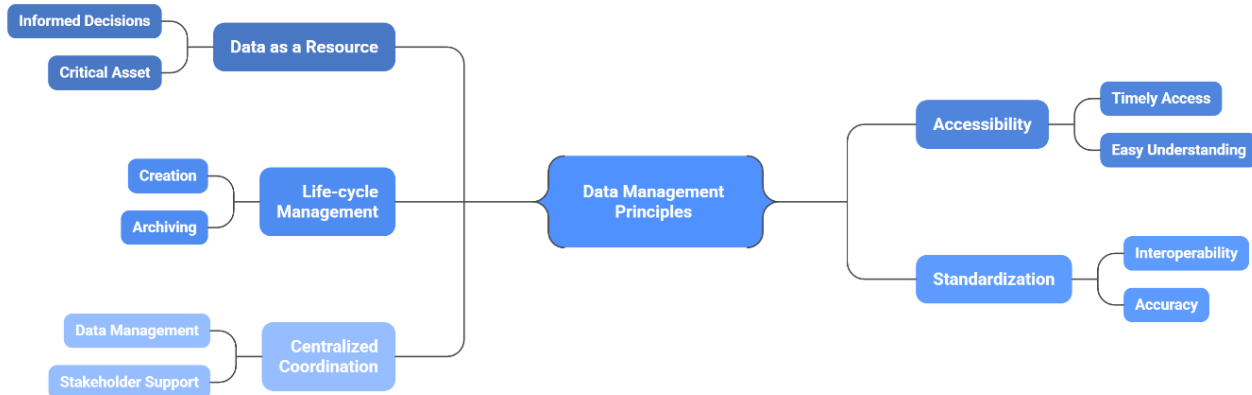


Figure 4 The Vision & Guiding Principles

Table 6 Key Components for Implementation

Category	Component	Key Activities (Summarized)
Technical Development	Robust Database Implementation	Select DBMS; define structures and metadata; implement backup, warehousing, and access hierarchy
	Hardware Consolidation	Consolidate equipment; install standby server; secure racks; document installations
	GIS Integration	Integrate GIS unit into MEIC; acquire GIS server/software; develop layers; ensure ISO compatibility
	Security Protocols	Implement firewall, antivirus, PKI; develop Security Policy; conduct IV&V; audit changes
Data Management	Define Core Data Elements	Develop indicators; prioritize collection; standardize structures; define provider obligations

	Data Validation & Verification	Apply automatic and manual validation; use batch totaling; verify completeness against licenses
	Metadata Repository	Create metadata structures; document core metadata; enable search and interoperability
Reporting & Dissemination	Streamlined Reporting	Align with EU and national laws; develop templates; produce State of Environment Reports
	"Report Once, Use Many Times" Model	Implement Reportnet principles; use XML; publish in accessible formats via website
Institutional & Human Factors	Building Cooperation	Establish steering committee and taskforces; hold workshops; engage donors and industry
	Motivation & Training	Conduct teambuilding workshops; provide IT and administrator training; organize study tour; establish helpdesk
	Organizational Change	Integrate GIS into MEIC; split MEIC into Data Management and IT units; hire additional staff

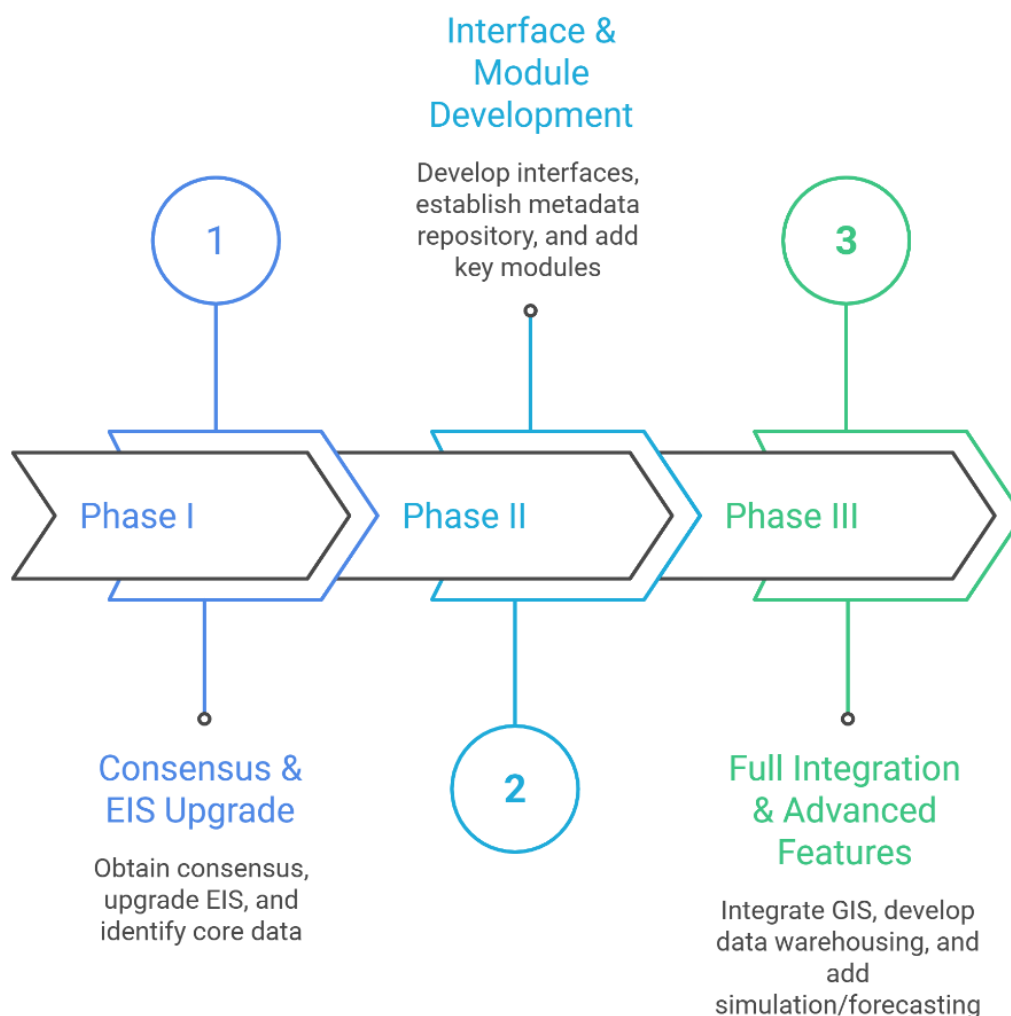


Figure 5 Three-phase of Strategy implementation

3.1.7 Sweden: Strategy for Environmental Data Management

Vision:

“Well managed environmental data is of significant benefit and makes a difference for the environment”.

The Swedish Strategy for Environmental Data Management (MIT, 2015) emphasizes that environmental data should be made available free of charge with permissive conditions to encourage use and reuse. The purpose of the strategy is to provide a framework for supporting government and agency to deal with environmental data as an important

community resource. It also aims to provide a framework for organisations to follow when integrating data management in an efficient manner.

Strategy main objective for environmental data management is to make optimal use of quality managed environmental data to benefit the environment. The environmental data should follow specific criteria including being: available, useable and managed efficiently in order to support environmental assessment, analysis, insights, decision-making and action-taking.



Figure 6 Sweden: Strategy for Environmental Data Management Guidelines

3.1.8 UK Environmental Data Service (EDS)

The UK Environmental Data Service (EDS) is a coordinated, distributed data service funded by the Natural Environment Research Council (NERC), UK Research and Innovation (UKRI). According to the NERC Environmental Data Service (2024), the EDS provides integrated data services across environmental science domains. It provides integrated data services across NERC's environmental data holdings. Data centres are working together through the EDS to share skills and technology in order to develop and provide world class data services. The EDS is a single interface providing access to scientific data and information across all environmental science disciplines. It supports the work of all those involved in environmental science including researchers, industry, government and the wider public (NERC Environmental Data Service, 2024).

Vision:

The EDS aims to make the long term availability, accessibility and re-use of environmental data possible as a sustainable national resource so that its full potential can be realised and the associated benefits can be realised and the associated benefits can be fully realised for the benefit of society in order to: Facilitate the easy search, access and use of the data to reveal new environmental knowledge that will have the potential to transform our understanding of environmental change.

Objectives:

The EDS will achieve a number of Key Objectives that are consistent with FAIR principles (Findable, Accessible, Interoperable, Reusable):

- 1 Deliver integrated data services across the breadth of environmental science, and move away from the current situation of separate silos for each discipline, in order to support truly interdisciplinary research and address the major, society-driving, environmental issues of the day.
- 2 It Informs researchers of required data preparation in order to archive and share their data appropriately. Ensure that data is of high enough quality to be archived and shared appropriately.
- 3 Develop common back-end services and front-end tools used to facilitate data discovery, access and analysis.
- 4 Development of data standards and interoperability frameworks including vocabularies and ontologies for disciplinary science and engagement of wider stakeholders.



Figure 7 UK NDS Key Services

4 Synergies between Kuwait Environmental Data Management Plan (NEDMP) objectives and Objectives of Key Practices

Analysis of the findings shows that the NEDMP objectives are well-aligned with international, regional, and national best practices, ensuring that Kuwait's approach benefits from global expertise while being tailored to national priorities.

Summary of Findings

1. **GEDS (UNEP Global Environmental Data Strategy)** is consistently strongly aligned with all five NEDMP objectives. This is reflecting the foundational role that the GEDS framework is playing in shaping Kuwait's national approach.
2. **ENPI-SEIS (Shared Environmental Information System)** and **Arab League Environmental Indicators** show strong alignment across most objectives, particularly highlighting the importance of regional cooperation and harmonization.
3. **Statistical Frameworks (FDES, SEEA)** are most strongly aligned with **SO2 (Data Quality)**, reflecting the important role in standardizing environmental data collection and reporting.
4. **WESR (UNEP World Environment Situation Room)** is strongly aligned with **SO3 (Interoperability)** and **SO4 (Data Sharing)**, reflecting its function to align Kuwait's national environmental data management with UN data platform.
5. **Country case studies (Macedonia, Sweden, UK)** provide valuable reference points, with the UK's Environmental Data Strategy showing strong alignment across multiple objectives, particularly for interoperability and data sharing.

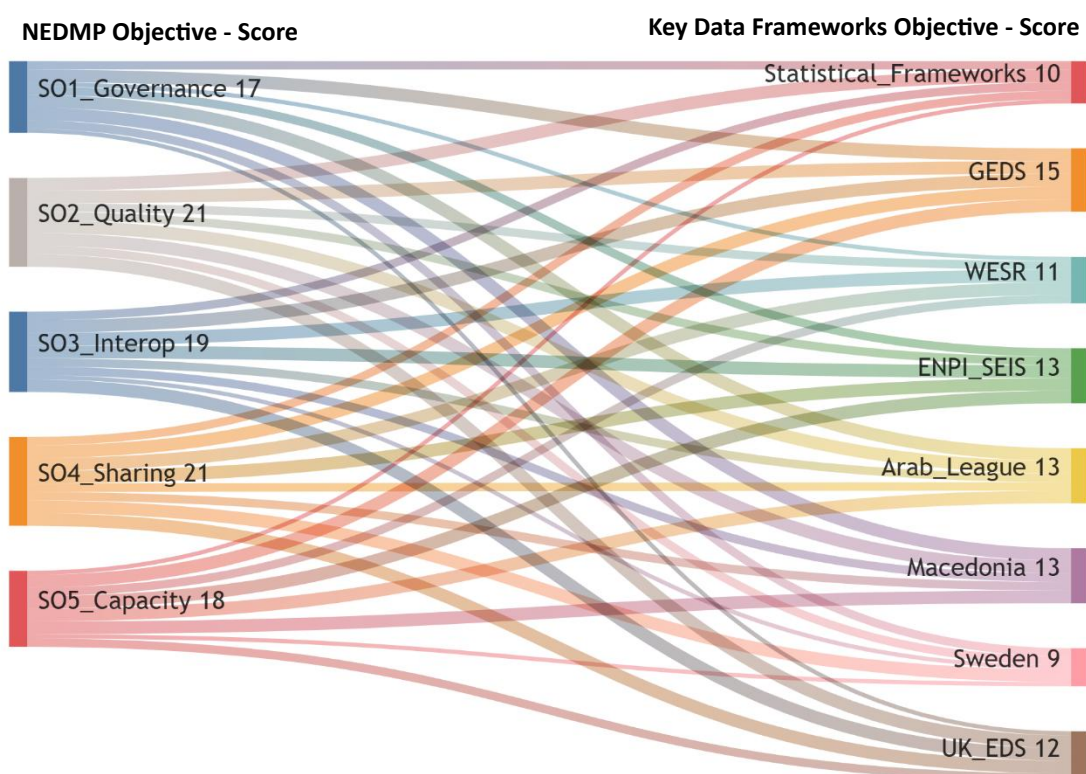


Figure 8 Synergies between (NEDMP) objectives and Objectives of Key Practices

Table 7 Diagram Label Key

Source Label	Full Name
SO1_Governance	SO1: Institutionalize Environmental Data Governance
SO2_Quality	SO2: Ensure High-Quality and Standardized Environmental Data
SO3_Interop	SO3: Enable Interoperability and System Integration
SO4_Sharing	SO4: Strengthen Data Sharing, Accessibility and Transparency
SO5_Capacity	SO5: Build Sustainable National Capacities
GEDS	UNEP Global Environmental Data Strategy
Statistical_Frameworks	Statistical Frameworks (e.g., FDES, SEEA, SDMX)
WESR	UNEP World Environment Situation Room
ENPI_SEIS	ENPI-SEIS (Shared Environmental Information System)
Arab_League	Arab League Environmental Indicators
Macedonia	North Macedonia – Environmental Data Management Practice
Sweden	Sweden – Environmental Data Governance Practice

Table 8 Summary of Environmental Data Management Practices

Case Study	Coverage	Type of Plan	Main Governance Entity	Components (Summarized)
The Global Environmental Data Strategy (GEDS)	Global	Strategy	United Nations Environment Programme (UNEP)	Pillars: Management, Quality & Provenance, Interoperability, Accessibility & Affordability, Capacity-Building
Environmental Data Management Statistical Frameworks	Global	Frameworks / Standards	United Nations Statistics Division (UNSD) and UNEP	FDES 2013, SEEA, UN Fundamental Principles of Official Statistics
World Environment Situation Room (WESR)	Global	Data & Knowledge System	United Nations Environment Programme (UNEP)	Geospatial, thematic, statistical, and near real-time data; SDG dashboards
ENPI-SEIS Project	Regional (ENP Region)	Data collection, Capacity-Building & Implementation Project	European Environment Agency (EEA) and EC services	Cooperation, content, infrastructure; priority themes: water, waste, air, climate change
Sustainable Development Indicators for the Arab Region	Regional (Arab Region)	Data and Indicator Harmonization Plan	League of Arab States (LAS)	Harmonized indicator package, structure, governance mechanism, priority themes, work plan
Republic of Macedonia: Environmental Data Management Strategy	National	Strategy & Implementation Plan	Ministry of Environment and Physical Planning (MEPP)	Technical development, data management, reporting & dissemination, institutional & human factors
Sweden: Strategy for Environmental Data Management	National	Strategy & Guidelines	The Council for Management and Development of Business Solutions (MIT)	Data availability, usability, efficient management for assessment and decision-making
UK Environmental Data Service (EDS)	National	Coordinated Data Service	Natural Environment Research Council (NERC) / UKRI	Integrated data services, data quality, common tools, data standards and interoperability

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